

Reflection on the Design Process

Battery Management System for Electric Scooter | ECE 445 Team 35

The design process for the battery management system taught me that a working embedded hardware project is not built by selecting individually correct parts, but by proving that the interfaces between those parts are electrically, thermally, and logically compatible. Early in the project, many design decisions looked straightforward at the block-diagram level: the LTC6811 would measure cell voltages and temperatures, the STM32 would process the data, the LTC6820 would handle isoSPI, and the power tree would generate the required rails. The important learning came when those blocks had to operate together on a real board.

One of the clearest examples was the master-board buck converter. The first design used a 4.7 μH inductor, which seemed reasonable from a component-size and availability perspective. However, doing the ripple-current calculation showed that the inductor current ripple was approximately 2.67 A at the worst-case 25.2 V input. That ripple was too large relative to the expected load current and pushed the converter toward discontinuous conduction mode. This made the 12 V rail less stable and affected the downstream 5 V and 3.3 V regulation. Increasing the inductor to 10 μH reduced ripple to about 1.25 A and made the design much more stable. From this, I learned that power design cannot be validated only by nominal voltage conversion; conduction mode, ripple, load range, and downstream sensitivity all matter.

The slave board also showed how electrical and thermal design are connected. Passive balancing was simple conceptually, but the bleed resistor dissipated about 1.76 W at 4.2 V with a 10 ohm resistor. That amount of localized heat can affect nearby MOSFETs and measurement traces if layout is not considered carefully. Adding thermal vias, copper area, and better component placement became just as important as the schematic itself.

Overall, the process taught me to treat calculations, layout, firmware behavior, and verification as one design loop. A requirement is only meaningful when it is tied to a measurable test, and a circuit is only complete when its non-ideal behavior has been considered. I also learned to document assumptions early, because values such as pack voltage range, load current, switching frequency, ADC range, and thermal limits directly determine whether later debugging is systematic or mostly guesswork.